10.10.3 Bonding in some Heteronucleardi-atomic molecules

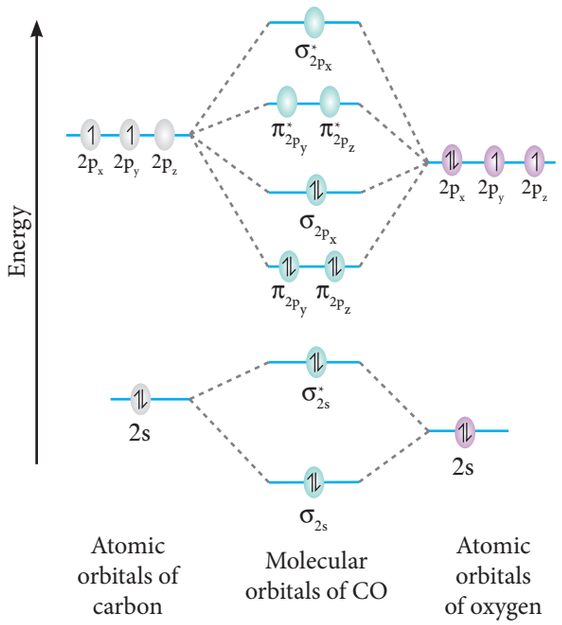


Fig 10.36 MO Diagram for CO molecule

Molecular orbital diagram of Carbon monoxide molecule (CO)

Electronic configuration of C atom is **$1s^2\,2s^2\,2p^2$**

Electronic configuration of O atom is **$1s^2\,2s^2\,2p^4$**

Electronic configuration of CO molecule $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi^2_{2p_y}, \pi^2_{2p_z}, \sigma^2_{2p_x}$

Bond order $= \frac{N_b - N_a}{2} = \frac{10-4}{2} = 3$

Molecule has *no unpaired electrons*. Hence, it is **diamagnetic**.

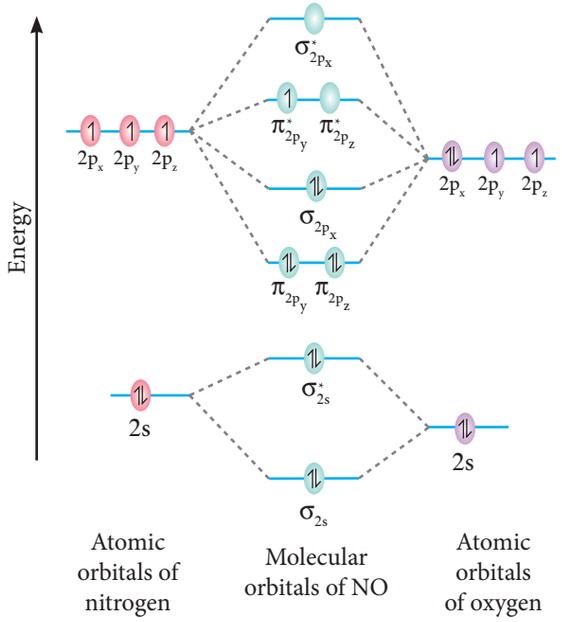


Fig 10.36 MO Diagram for NO molecule

Molecular orbital diagram of Nitric oxide molecule (NO)

Electronic configuration of N atom is **$1s^2\,2s^2\,2p^3$**

Electronic configuration of O atom is **$1s^2\,2s^2\,2p^4$**

Electronic configuration of NO molecule $\sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi^2_{2p_y}, \pi^2_{2p_z}, \sigma^2_{2p_x}\ \pi^{*1}_{2p_y}$

Bond order $= \frac{N_b - N_a}{2} = \frac{10-5}{2} = 2.5$

Molecule has *one unpaired electron*. Hence, it is **paramagnetic**.

Evaluate Yourself

10) Draw the MO diagram for acetylide ion C_2^{2-} and calculate its bond order.

Unit 10 layout - jagan.indd 101 22-01-2021 12:33:44